

MAT3032 ADVANCED ALGEBRA: UNASSESSED COURSEWORK 1 SOLUTIONS

1. You can see the tables in Möbius/Maple TA. View your past results and the correct answers will be seen alongside your own.

- (a) $\mathbb{Z}_8$ has 1 element of order 2, namely 4.
- (b) $\mathbb{Z}_2 \times \mathbb{Z}_4$ has 3 elements of order 2, namely $(0, 2), (1, 0), (1, 2)$.
- (c) $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ has 7 elements of order 2: all except $(0, 0, 0)$.
- (d) This group is isomorphic to D_4 and has 5 elements of order 2: A^2, B, AB, A^2B, A^3B . (The corresponding elements of D_4 are the half-turn and four reflections of a square.)
- (e) This group is not isomorphic to D_4 and has 1 element of order 2.

It must therefore be isomorphic to Q_4 .

(Note that this is a different representation of Q_4 from the one in Exercises 1.2 Question 1. An isomorphism between the two is given (not uniquely) by

$$1 \mapsto 1, \quad A \mapsto i, \quad A^2 \mapsto -1, \quad A^3 \mapsto -i,$$

$$B \mapsto j, \quad AB \mapsto k, \quad A^2B \mapsto -j, \quad A^3B \mapsto -k.$$

where A and B are as on Page 6 of the notes, not as in 1(d) above.)

- 2. (a) $\varphi(15) = 15(1 - 1/3)(1 - 1/5) = 8, \quad \varphi(16) = 16(1 - 1/32) = 8,$
 $\varphi(20) = 20(1 - 1/2)(1 - 1/5) = 8, \quad \varphi(24) = 24(1 - 1/2)(1 - 1/3) = 8,$
 $\varphi(30) = 30(1 - 1/2)(1 - 1/3)(1 - 1/5) = 8.$
- (b) The numbers of elements of order 2 are respectively 3, 3, 3, 7, 3 so the groups of units are all isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_4$ except U_{24} which is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.

- 3. (a) $I_3 \in G$ so G is non-empty.

The elements of G are non-singular, as they have determinant 1, so G is a subset of $GL(3, \mathbb{Z}_p)$.

$$\text{Let } X = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}, Y = \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix} \text{ be in } G.$$

$$\text{Then } XY = \begin{pmatrix} 1 & a+d & b+e+af \\ 0 & 1 & c+f \\ 0 & 0 & 1 \end{pmatrix} \text{ (all calculations being modulo } p).$$

This is clearly in G . As G is finite, this completes the subgroup test, showing that G is a subgroup of $GL(3, \mathbb{Z}_p)$.

$$YZ = \begin{pmatrix} 1 & a+d & b+e+cd \\ 0 & 1 & c+f \\ 0 & 0 & 1 \end{pmatrix} \neq XY \text{ in general, so } G \text{ is non-abelian.}$$

Each of a, b, c can be chosen in p ways (any of $0, 1, \dots, p-1$), so $|G| = p^3$.

- (b) For X, Y as above, $\psi(X) = a + c$, $\psi(Y) = d + f$
 and $\psi(XY) = (a + d) + (c + f) = (a + c) + (d + f) = \psi(X) + \psi(Y)$.
 Thus ψ is a group homomorphism from $(G, \times)$ to $(\mathbb{Z}_p, +_p)$.
 $\text{Ker } \psi$ consists of all matrices X in which $a + c = 0$.
 Thus a, b can each be chosen freely, with any of p values each, but c must equal $-a$ (or equivalently $p - a$), so $|\text{Ker } \psi| = p^2$.
 The image of ψ is $\mathbb{Z}_p$, since every element of $\mathbb{Z}_p$ can be obtained by applying ψ to some matrix in G . (e.g. to get $\psi(X) = q$, put $a = q, c = 0$ in X .)
 Thus $|\psi(G)| = |\mathbb{Z}_p| = p$. (ψ is surjective but not injective.)
- (c) $\text{Ker } \phi$ is a normal subgroup of G (the kernel of a homomorphism is always normal in the domain) so the left and right cosets are the same. (The number of cosets is the index of $\text{Ker } \phi$ in G , i.e. p^3/p^2 , which equals p .)
 X and Y , as above, are in the same coset of $\text{Ker } \psi$ iff $X \text{Ker } \phi = Y \text{Ker } \phi$, i.e. $X^{-1}Y \in \text{Ker } \phi$. This holds iff $\phi(X^{-1}Y) = 0$, i.e. $-\phi(X) + \phi(Y) = 0$ (since ϕ is mapping from a multiplicative to an additive group).
 Thus we need $\phi(X) = \phi(Y)$ so the condition is $a + c = d + f$, or equivalently $a + c - d - f = 0$, in $\mathbb{Z}_p$.
- (d) There are three non-isomorphic *abelian* groups of order p^3 , namely C_{p^3} , $C_p \times C_{p^2}$ and $C_p \times C_p \times C_p$. Together with the non-abelian group G in this question, there are at least four distinct groups of order p^3 .
 Of these, only C_{p^3} is cyclic. (All cyclic groups of any given order are isomorphic.)
- (e) Taking $p = 3$, the website given in the question tells us that there are actually 5 groups of order 27. (So there is one more that we haven't seen.)

4. (a) $\tau\sigma\tau^{-1}(1) = \tau\sigma(1) = \tau(4) = 5$, etc,
 giving $\tau\sigma\tau^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \end{pmatrix} = (1 \ 5 \ 4 \ 3 \ 2)$.
 This is a product of four transpositions, $(1 \ 2)(1 \ 3)(1 \ 4)(1 \ 5)$, so it is in A_5 .
- (b) Similarly, $\tau\sigma\tau^{-1} = (1 \ 5 \ 4)(2 \ 3)$. This is a product of three transpositions, $(1 \ 4)(1 \ 5)(2 \ 5)$, so it is not in A_5 .
- (c) $\tau\sigma\tau^{-1} = (1 \ 2 \ 3 \ 5)$. This is a product of three transpositions, $(1 \ 5)(1 \ 3)(1 \ 2)$, so it is not in A_5 .

Note that in each part, conjugating σ by τ gives a permutation in which the disjoint cycles have the same lengths as in σ . We shall prove in Section 2.3 that this is always the case.

5. The number of distinguishable colourings with all four triangles the same colour is 3 (all blue, all green or all red).

The number with the three corner triangles the same colour and the centre one different is 6 (three choices for the corners; for each of these, two choices for the centre).

The number with a corner triangle in one colour and the other three in a second colour is 6 (three choices for the corner; for each of these, two choices for the other three).

The number with two triangles in one colour and two in another colour is 6. (three choices for first two; for each of these, two choices for the other two).

The number with two corner triangles the same and the other two triangles different is 6. (three choices for the two corners; for each of these, two ways to colour the other two).

The number with a corner triangle and the centre triangle the same and the other two different is 3 (three choices for the corners and centre; only one distinguishable way to colour the other two, as the triangle can be viewed from both sides).

The total number of distinct colourings is 30.

The point of doing this the hard way is so that you will appreciate the much easier method that will be described in Section 2.4.